

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Understand basic data structures using AI assistance
- Implement stack, queue, and linked list
- Analyze operations and efficiency
- Improve coding practices using AI suggestions

Task 1: Stack Implementation

Python Code

```
stack = []

stack.append(10)
stack.append(20)
stack.append(30)

print(stack.pop())
print(stack)
```

Explanation: Stack follows LIFO (Last In First Out).

Task 2: Queue Implementation

Python Code

```
from collections import deque

queue = deque()

queue.append(10)
queue.append(20)
queue.append(30)

print(queue.popleft())
print(queue)
```

Explanation: Queue follows FIFO (First In First Out).

Task 3: Linked List Implementation

Python Code

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
second = Node(20)
third = Node(30)

head.next = second
second.next = third

temp = head
while temp:
    print(temp.data, end=" ")
    temp = temp.next
```

Explanation: Linked list stores elements in nodes connected via pointers.

Task 4: Operations Analysis

- Stack: Push and Pop are $O(1)$
- Queue: Enqueue and Dequeue are $O(1)$
- Linked List: Traversal is $O(n)$

Conclusion

AI tools simplify implementation of data structures.

Understanding underlying concepts is essential for efficient problem solving.